Homework: GDP of the Kingdom of Saudi Arabia

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Country and Data Selection

How accurately can we forecast GDP for an oil economy, and what if that economy has only begun to provide complete monthly data over the past fifteen years? Our project addresses these questions by forecasting Saudi Arabia's GDP. We construct a dynamic factor model in which a latent monthly factor summarizes the common movement across Saudi macroeconomic indicators, and we use the model to forecast quarterly real GDP growth for the next two quarters.

The sample covers January 2015 to June 2026. The dataset combines quarterly real GDP with monthly indicators of oil and non-oil activity, reflecting the structure of the Saudi economy. Crude oil prices and exports capture oil-sector activity, while imports and point-of-sale transactions proxy domestic demand and investment-related activity.

For our extension, we change the target from headline GDP to non-oil GDP. Because soft indicators and financial variables did not improve the baseline model for aggregate GDP, which is heavily influenced by oil-sector fluctuations, we hypothesize that these variables may be more informative about the non-oil economy. We therefore add PMI, quarterly financial BOP, and the Tadawul All Share Index (TASI), and remove exports. The resulting factor exhibits a substantially stronger correlation with its target series, increasing from 0.53 to 0.68.

Table 1: **Variables used to construct the common factor (** = Extension-only variables)

Variable	Frequency	Economic interpretation
Real GDP*	Q	Target variable
Foreign Reserves	M	External position and foreign asset accumulation
Goods Exports*	M	Oil-sector activity and external demand
Goods Imports	M	Domestic demand and investment activity
Brent Crude Oil Price	M	Global oil-market conditions
Point-of-Sale Transactions	M	Household consumption
Real Non-Oil GDP	Q	Target variable
PMI	M	Business sentiment and expected activity
Tadawul All Share Index (TASI)	M	Financial conditions and market expectations
Financial Balance of Payments	Q	Capital flows and financial-sector activity

Notes: Variables highlighted in blue are included only in the extension specification. Variables marked with * are omitted from the extension model. Data are obtained from GASTAT, FRED, IMF International Financial Statistics, the U.S. Energy Information Administration, JODI Oil, and the Saudi Central Bank (SAMA). All monthly variables are transformed as simple growth rates, $\frac{x_t}{x_{t-1}} - 1$. GDP is measured as quarterly log growth. PMI is included in levels in the extension specification.

Results

Table 2: Model comparison and forecasts

Statistic	Baseline Model	Extension Model
Target variable	Real GDP	Non-oil GDP
Factor correlation to target	0.533	0.678
Q+1 forecast growth	1.34 %	1.25 %
Q+2 forecast growth	0.77 %	1.16 %

Validation Plots

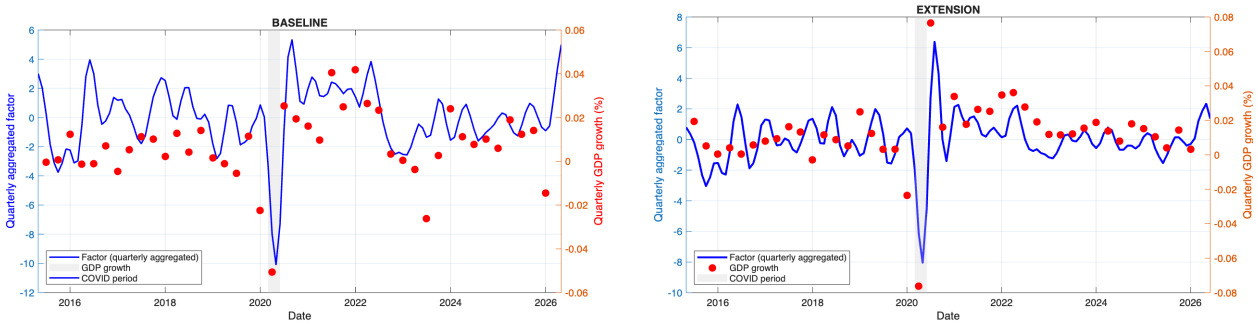


Figure 1: Estimated factors and observed GDP growth. The left panel shows the baseline factor against quarterly real GDP growth, while the right panel shows the extended factor against quarterly non-oil GDP growth. The extension model exhibits stronger co-movement with the target series (0.53 vs. 0.67).

Discussion

- **Headline GDP is more volatile than non-oil GDP, and more difficult to forecast**. The comparison of the quarterly GDP series in the above plots highlights how oil-sector dynamics contribute substantially to the volatility of headline GDP, while non-oil GDP contains less extreme oscillations. Composing a less oil-oriented factor also smoothens factor's trajectory, and the correlation between the two jumps to .68. Financial variables such as the TASI index and the balance of payments, together with PMI data, are also more forward-looking and therefore improve forecasts of non-oil GDP. Furthermore, the extension not only increases the factor's correlation with the target series (0.53 to 0.68), but also substantially improves the model's internal fit (0.56 to 0.90), suggesting that the added financial and sentiment indicators capture a larger share of the common variation in Saudi economic activity.
- **Oil-related variables provide limited information about future activity.** Saudi oil revenues depend on global oil prices and OPEC+ production decisions, both of which are largely exogenous to the domestic Saudi economy and highly volatile. While these variables explain a substantial share of contemporaneous movements in headline GDP, they add noise to the forecasting problem because future fluctuations are often driven by external geopolitical and market developments rather than domestic economic conditions. The Saudi economy is also structured to partially offset movements in prices and production, further limiting the signal these variables can provide.
- **Data limitations constrain the scope of the analysis.** Since the common factor is constructed only from data available from 2015 onward, the sample period is relatively short, preventing the implementation of more data-intensive methodologies such as Markov-Switching models.
- **Recent geopolitical developments:** The model may not fully assess the impact of recent geopolitical developments, including the US–Israel conflict with Iran, as some of the available oil-related data do not extend far enough to capture these events and their macroeconomic consequences. Despite recent geopolitical tensions, neither specification currently predicts a contraction in Saudi GDP over the forecast horizon. This result should be interpreted cautiously, as the latest data may not yet fully capture the economic effects of the conflict.